

Botond Branyicskai-Nagy

London, United Kingdom

botondbnagy@gmail.com

Education

University College London

September 2023 – September 2024

MSc Machine Learning (*Distinction*)

Probabilistic and Unsupervised Learning¹ | Approximate Inference¹ | Bayesian Deep Learning
Applied ML | Supervised Learning | Reinforcement Learning² | Applied Deep Learning

MSc Thesis (*Distinction*):

Hierarchical Bayesian Program Synthesis for Neural Algorithmic Reasoning: wake-sleep Bayesian algorithm learning with neural library of primitives, and a transformer-based generative model for synthesis. Supervised by Prof. Mirco Musolesi.

Imperial College London

October 2020 – June 2023

BSc Physics with Theoretical Physics

BSc Thesis (*Distinction*):

Computational Evolution: Simulation and analysis of evolutionary dynamics. Comparison with theoretical models, exploration of optimisation algorithms. Supervised by David Clements.

Experience

ML Research Consultant – Stealth Startup (Consulting)

2025

Conducted a feasibility study on integrating Large Language Models (LLMs) into the company's core product. Delivered a proof of concept implementation using OpenAI's API for interpreting surveys and generating schemas. Developed and tested competing NLP pipelines, authoring detailed methodology, implementation guides, and feasibility recommendations.

Visiting Researcher — University College London

2024

Research on Neural Program Synthesis and Algorithmic Reasoning in the Machine Intelligence Lab, supervised by Prof. Mirco Musolesi.

Research Intern — Imperial College London

2023

Causal discovery using cross-validation as compared with Bayesian Model Selection (GPLVM), supervised by Mark van der Wilk and funded by Imperial UROP.

Projects

Masked Image Modelling — University College London

2024

Self-supervised learning framework: a VisionTransformer is pre-trained to reconstruct partially masked images then fine-tuned on the Oxford-IIIT Pets dataset for segmentation. (*Distinction*)

Solar Sail (Project Svarog) — Imperial College Space Society

2022

Orbital mechanics: spacecraft trajectory simulation, atmospheric effects, relativistic corrections.

Machine learning for LHCb experiment — Imperial College London

2022

Leading ML team in project analysing decay product distribution from LHCb experiment for Standard Model comparison. Neural Networks (TensorFlow), Decision Trees. (*Distinction*)

Laser microphone — Imperial College London

2020

Building a proof-of-concept prototype of a laser interferometer-based device capable of recovering audio from vibrations of a window from a distance. (*Distinction*)

Awards

2nd Year Quantum Physics Article Prize – Published on Imperial Website (2022)

Young Researchers of Natural Sciences, Conference Presentation Distinction (2016)

Skills and Interests

Languages & Programming

Python - Scientific and ML libraries (PyTorch, TensorFlow, GPflow, JAX), Julia, Cluster usage

Languages

Hungarian (Native), English (Proficient - Cambridge Advanced 206), French (Intermediate)

¹Module part of PhD programme at the Gatsby Computational Neuroscience Unit

²Module taught by Google DeepMind

For references please contact:

Dr Mark van der Wilk

Associate Professor,
Department of Computer Science,
University of Oxford
`mark.vdwilk@cs.ox.ac.uk`

Professor Daniel Waldram

Professor of Theoretical Physics,
Head of the Theoretical Physics Group,
Department of Physics,
Imperial College London
`d.waldram@imperial.ac.uk`